

ASSESSMENT OF SURFACE WATER MICROBIOLOGICAL QUALITY AND PROSPECTIVE TREATMENT FOR CONTAMINANT REMOVAL

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Abstract

Surface water quality is influenced by natural geochemical processes, urban activities, industrial discharges, agricultural runoff and microbiological contamination. The present study evaluates the physicochemical, heavy-metal, microbiological and organic-pollutant characteristics of four surface-water samples collected from the investigated study area. The analyzed parameters included pH, total dissolved solids (TDS), total hardness, chloride, sulphate, turbidity, colour, fluoride, iron, aluminium, lead, cadmium, manganese, total coliform and *Escherichia coli*. Total coliform was detected in all four samples, with Sample 3 recording the highest concentration of 80 MPN/100 mL, whereas *E. coli* was reported as <2 MPN/100 mL in all samples. Pesticides, PCBs, PAHs, DDT derivatives, endosulfan compounds, aldrin, dieldrin and related organic contaminants were reported below detection limits. The results indicate that the investigated surface-water samples are not suitable for direct drinking without appropriate treatment. The proposed treatment approach should be experimentally validated through analysis and post-treatment water-quality assessment.

Keywords: Surface water; water quality; salinity; turbidity; microbiological contamination;

How to cite this article: Janarthanan P Sheena A D. ASSESSMENT OF SURFACE WATER MICROBIOLOGICAL. Int J Drug Deliv Technol. 2026;16(82s):574-578. DOI: 10.25258/ijddt.16.82s.66.

1. Introduction

Surface-water resources are important for domestic, agricultural, industrial and ecological applications. However, their quality can deteriorate because of urbanization, industrial activities, agricultural runoff, wastewater discharge, erosion, mineral dissolution and microbiological contamination. Continuous monitoring of surface-water quality is therefore essential for determining its suitability for intended uses and identifying the major contaminants requiring treatment. Physicochemical parameters such as pH, TDS, hardness, chloride, sulphate, turbidity and colour provide important information about the general condition of surface water. High TDS and chloride concentrations indicate substantial dissolved ionic content, while elevated hardness reflects the presence of calcium and magnesium salts. Turbidity and colour are important indicators of suspended and dissolved materials and can affect the aesthetic and treatment characteristics of water. Heavy metals are of particular concern because some metals can persist in aquatic environments and accumulate in biological systems. Lead and cadmium are especially important

contaminants because of their toxicity and potential human-health effects. Iron, manganese and aluminium may also deteriorate water quality when present at elevated concentrations. Microbiological contamination represents another important dimension of surface-water quality. The occurrence of total coliform organisms indicates possible contamination from environmental and anthropogenic sources. Although the absence of detectable *E. coli* provides useful information, it does not necessarily establish complete microbiological safety.

Microbiological assessment shows presence of total coliforms in all the four sampling points of Kosasthalaiyar River under study with concentration varying between 6 and 80 MPN/100 mL. Sample S3 showed the highest concentration (80 MPN/100 mL) and pointed out a hot spot of localized microbiological contamination. Total coliforms in surface water may be associated with sewage discharges, domestic wastewater, animal wastes, surface runoff and other environmental sources; however, total coliforms

should not be considered as conclusive evidence of fecal contamination alone as a number of members within the group may be found naturally occurring in aquatic environments (Edberg et al., 2000; World Health Organization, 2022). In the current study *E. coli* was reported as <2 MPN/100 mL in all four samples indicating that detectable *E. coli* contamination was not detected at the analytical reporting limit used. In 2020, an earlier environmental assessment of the Chennai Peripheral Ring Road recorded 23 MPN/100 mL total coliform and 12 MPN/100 mL *E. coli* at a sampling point in the Kosasthalaiyar River (Chennai Peripheral Ring Road EIA, 2020). The present maximum concentration at S3 (80 MPN/100 mL) surpasses the total coliform value reported earlier; however, the difference should not be interpreted as a temporal increase as the sampling locations, dates, seasons, hydrological conditions and analytical circumstances may differ. CPCB (2008) has set the limit of ≤ 50 MPN/100 mL for total coliform for Class A water and ≤ 500 MPN/100 mL for Class B water under the Designated Best Use criteria. S3 therefore surpassed the Class A criterion, and S1, S2 and S4 fell below the criterion. However, compliance with the total-coliform criterion alone does not indicate drinking-water suitability, which requires consideration of physicochemical, chemical and microbiological parameters together. Earlier studies on Indian river systems have also shown large spatial and seasonal variation in microbial contamination and associated high indicator bacteria with settlement, sewage related inputs and surface runoff (Pradhan et al., 2016). Therefore, the present findings are best interpreted as evidence of localized microbiological stress on water quality rather than definitive evidence of fecal pollution or a basin-wide temporal deterioration. However, the results support the need for continued spatial and seasonal monitoring and incorporation of an appropriate disinfection barrier into any treatment system intended for domestic or potable applications (World Health Organization, 2022).

Materials and Methods

Surface water samples were collected from selected locations in the investigated study area to examine the overall physicochemical, heavy-metal and microbiological quality of the water. Four representative sampling locations were selected taking into account differences in land use, anthropogenic activity, drainage characteristics and potential sources of contamination. Sampling was carried out with clean sterilized sampling containers suitable for the respective analysis.

Samples were collected in clean polyethylene bottles for physicochemical and chemical analysis and in sterile contamination free bottles for microbiological analysis. The sampling containers were labeled properly with the site of sampling, date and the identification number of the sample. Microbiological samples were collected aseptically, avoiding contact with the inner surface of the container or its cap. The collected samples were immediately transported to the laboratory under cooled conditions and analyzed within the recommended holding period. Samples were checked for visual color, odor and suspended matter prior to laboratory analysis. Physicochemical analysis samples were stored under controlled laboratory conditions before analysis. Samples for heavy-metal determination were properly preserved to prevent adsorption, precipitation and other changes during storage. Microbiological samples were transported under refrigerated conditions and processed as soon as possible after collection. All sampling and glassware for microbiological analysis was autoclaved before use. All the reagents were of analytical grade and de-ionized water was used in the whole investigation. Standard laboratory procedures were followed for the preparation of reagents, calibration of instruments and analysis of the selected water quality parameters.

3.1 Surface-Water Sampling

Four surface-water samples designated S1, S2, S3 and S4 were considered for the present investigation. Samples were collected from the selected surface-water locations within the study area and subjected to physicochemical, chemical and microbiological analysis.

The samples were analyzed for major water-quality parameters, heavy metals, microbiological indicators and selected organic contaminants.

ANALYSIS OF STUDY

Microbiological quality was evaluated using **total coliform** and *Escherichia coli* as indicator organisms. The results obtained for the four surface-water samples are presented below.

Table X. Microbiological Quality of Surface-Water Samples

Parameter	Standard	S1	S2	S3	S4
<i>E. coli</i> (MPN/100 mL)	0	<2	<2	<2	<2

Total Coliform (MPN/100 mL)	0	6	7	80	9
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E. coli was reported as <2 MPN/100 mL in all four surface-water samples. However, total coliform bacteria were detected in every sample. Sample 3 recorded the highest total-coliform concentration of **80 MPN/100 mL**, substantially higher than the other sampling locations. This indicates a localized microbiological contamination hotspot. Although *E. coli* was not detected above the reported analytical threshold, the presence of total coliforms indicates that the surface water should not be considered microbiologically safe for direct consumption without appropriate treatment and disinfection.

3.5 Organic Pollutants and Pesticides

The investigated surface-water samples were analyzed for selected pesticides and organic pollutants, including PCBs, PAHs, DDT derivatives, endosulfan compounds, aldrin, dieldrin and related organic contaminants. All analyzed pesticides, PCBs, PAHs, DDT derivatives, endosulfan compounds, aldrin, dieldrin and related organic pollutants were reported as **Below Detection Limit (BDL)**.

Interpretation

The absence of detectable concentrations of the analyzed organic pollutants indicates that these compounds were not detected at or above the analytical detection limits in the investigated surface-water samples. Therefore, based on the available analytical results, detectable pesticide and persistent organic pollutant contamination was not a dominant component of the observed surface-water quality deterioration.

The BDL should be interpreted according to the analytical detection limit of each method. It does not necessarily indicate absolute absence of the compound.

RESULTS AND DISCUSSION

3.6 Surface-Water Quality Assessment

Table X. Overall Assessment of Surface-Water Samples

Sample	Assessment
Sample 1	Extremely mineralized and highly saline surface water

Sample 2	High turbidity, colour and iron contamination
Sample 3	Highest microbiological and selected heavy-metal contamination
Sample 4	Severe salinity and hardness contamination

The overall assessment demonstrates that the four samples exhibit different contamination profiles. Sample 1 is dominated by dissolved mineral contamination, Sample 2 by turbidity, colour and iron, Sample 3 by microbiological contamination and selected heavy metals, and Sample 4 by salinity, hardness and lead.

5.7 Microbiological Contamination

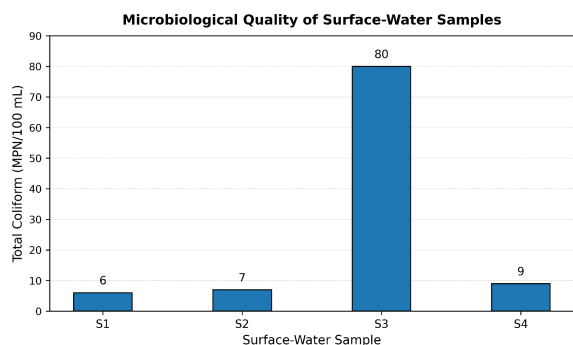
The total-coliform concentration ranged from **6 to 80 MPN/100 mL**.

Sample 3 recorded the highest concentration of **80 MPN/100 mL**, which was substantially higher than the concentrations observed in S1, S2 and S4. The detection of total coliform in all four samples indicates microbiological contamination of the investigated surface-water system. Consequently, any proposed treatment system intended to produce potable water should incorporate an appropriate disinfection stage.

5.8 Organic Pollutants

All analyzed pesticides and selected persistent organic pollutants were reported as BDL.

This indicates that detectable organic contamination was not a major component of the measured water-quality deterioration. The dominant observed problems were instead associated with dissolved salts, hardness, turbidity, colour, selected heavy metals and microbiological contamination.



Pesticide and Organic Pollutants Contamination

The analyzed surface-water samples were screened for a number of selected pesticides and persistent organic pollutants (POPs) such as polychlorinated biphenyls (PCBs), polycyclic aromatic hydrocarbons (PAHs), DDT and its derivatives, endosulfan compounds, aldrin and dieldrin. All analyzed compounds were reported as below respective analytical detection limits (BDL) for tested samples. The BDL results indicate that none of the organic contaminants studied were detected at or above their respective analytical detection limits when using the analytical conditions described. Thus, within the scope and sensitivity of the analytical procedures used, detectable pesticide and selected persistent organic-pollutant contamination was not a major contributor to the observed decline in surface-water quality. This result is particularly significant when considered together with the physicochemical and inorganic contaminant results. The main water quality problems in the present study were associated with high dissolved solids, hardness, chloride, turbidity, color, some heavy metals and microbiological indicators. Hence, the observed degradation appears to be more associated with the dissolved inorganic constituents, suspended matter, metal contamination and microbial indicators rather than the organic pollutants selected and studied.

The BDL should not be interpreted as evidence for the absolute absence of corresponding compounds. A BDL result signifies that the concentration of the specific compound was less than the analytical detection capability for the particular compound and method. Differences in extraction efficiency, matrix effects, analytical sensitivity and compound-specific detection limits may affect the reported result. Thus, the organic-pollutant results should be interpreted against the individual method detection limits (MDLs/LODs). Reporting the numerical detection limit for each analyte would be a further improvement

of the analytical transparency of the study.

Overall, the screening for organic contaminants gave valuable evidence that the selected pesticides and POPs were absent in the samples investigated in measurable quantities, while the more prominent water-quality issues were related to physicochemical, inorganic and microbiological parameters.

Comparison of Your Results with Published Kosasthalaiyar Data

Study	Water body	Total coliform (MPN/100 mL)	Interpretation
Present study – S1	Kosasthalaiyar River	6	Detected, relatively low
Present study – S2	Kosasthalaiyar River	7	Detected, relatively low
Present study – S3	Kosasthalaiyar River	80	Highest; microbiological hotspot
Present study – S4	Kosasthalaiyar River	9	Detected, relatively low
CPRR EIA (2020)	Kosasthalaiyar River	23	Microbial indicator detected

Proposed Treatment Mechanism

The proposed mechanism can be represented as:

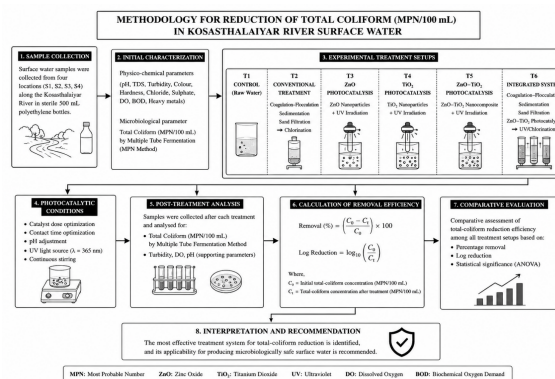


Fig. Propsed study for removal microbial polutants**CONCLUSION**

The EIA value of 23 MPN/100 mL in 2020 is particularly useful as a local supporting reference. However, I would not write that your study shows that the river is now more contaminated as the sampling conditions and locations are not necessarily the same. (Early Warning System).

Microbiological assessment showed that total coliforms were present in all the Kosasthalaiyar River sampling sites studied and ranged from 6 to 80 MPN/100 mL. Sample S3 showed a steep increase to 80 MPN/100 mL which suggested a localized hot spot of microbiological contamination. Independent surface-water monitoring for the Chennai Peripheral Ring Road EIA reported total-coliform concentrations of 23 MPN/100 mL in the Kosasthalaiyar River in 2020, corroborating the presence of microbial indicator organisms within this river system previously. (ES) The current S3 concentration is higher than this previously reported value, but the difference should not be interpreted as a temporal increase without an equivalent sampling location and seasonal observations. The CPCB designated-best-use criterion for Class A water is a total-coliform concentration of ≤ 50 MPN/100 mL. S3 is above the designated-best-use criterion for Class A Water while S1, S2 and S4 are below the designated-best-use criterion for Class A Water. (Central Pollution Control Board) However, total coliform alone cannot be used to assess microbiological quality and the occurrence of total coliforms should be interpreted along with *E. coli*, physicochemical parameters and other indicators of anthropogenic pollution.

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